Journal of Intelligent Systems and Control
https://www.acadlore.com/journals/JISC

Explainable Policy Distillation From Deep Q-Networks Into Compact Neuro-Fuzzy Controllers for Uncertain Nonlinear Systems

Mohammed Mansour[1*], Mustafa Kutlu[1], Turker Berk Donmez[2]

[1] Department of Mechatronics Engineering, Sakarya University of Applied Sciences, 54050 Serdivan, Turkey
[2] Department of Biomedical Engineering, Sakarya University of Applied Sciences, 54050 Serdivan, Turkey

[*] Correspondence: Mohammed Mansour (mohammedmansour@subu.edu.tr)

Received: 04-29-2025 **Revised:** 06-10-2025 **Accepted:** 06-24-2025

Citation: M. Mansour, M. Kutlu, and T. B. Donmez, "Explainable policy distillation from deep Q-networks into compact neuro-fuzzy controllers for uncertain nonlinear systems," *J. Intell Syst. Control*, vol. 4, no. 2, pp. 135–148, 2025. https://doi.org/10.56578/jisc040205.

© 2025 by the author(s). Licensee Acadlore Publishing Services Limited, Hong Kong. This article can be downloaded for free, and reused and quoted with a citation of the original published version, under the CC BY 4.0 license.

Abstract: A fundamental challenge in reinforcement learning-based control is the development of interpretable policies that retain effective control performance under system uncertainty and nonlinear dynamics. To address this challenge, a domain-randomized deep Q-network (DQN) policy was distilled into a compact Takagi–Sugeno neuro-fuzzy Q-controller for reference tracking in an uncertain nonlinear mass–spring–damper system. A fixed-premise fuzzy inference structure was adopted, in which rule activation strengths and consequent parameters were used to compute action values directly, thereby enabling explicit examination of the learned decision mechanism without an additional opaque decision head. The proposed distillation framework was evaluated against a directly trained neuro-fuzzy controller with an identical architecture. Closed-loop tracking performance, teacher–student action agreement, and rule-pruning fidelity were systematically assessed under nominal, robust, and stress conditions, using ten independent random seeds and 160 matched evaluation scenarios per operating regime. For the distilled controller with seven membership functions (7MFs) per input, the mean absolute tracking errors were 0.1589, 0.1552, and 0.3437, respectively, compared with 0.4989, 0.4299, and 0.5690 for its directly trained counterpart. Teacher–student action agreement decreased from under nominal conditions to under stress conditions. Rule-pruning analysis further demonstrated that a reduced controller retaining 64 of the original 343 fuzzy rules preserved of the full student's action selections. State-space and trajectory-level analyses were conducted to characterize policy disagreement and its observed association with tracking degradation. Although the distilled neuro-fuzzy controller outperformed a manually designed fuzzy controller, its tracking accuracy remained lower than that achieved by proportional–integral–derivative (PID) control, linear quadratic regulator (LQR) control, and the original DQN teacher. The resulting framework provides a basis for evaluating explainable policy compression in uncertain nonlinear control systems without implying guaranteed closed-loop stability or superior tracking performance.

Keywords: Explainable reinforcement learning; Policy distillation; Neuro-fuzzy control; Deep Q-networks; Uncertain nonlinear control

1 Introduction

Deep reinforcement learning can learn feedback policies for uncertain nonlinear dynamics, and domain randomization exposes policies to variations in system parameters during training. Generalization and safety, nevertheless, remain open under distribution shift [1–4]. A separate deployment obstacle is that neural action values do not explain which measured quantities drive a decision. Reviews of explainable reinforcement learning distinguish an intrinsically readable policy from a post-hoc account of an opaque one [5, 6].

Fuzzy controllers can expose antecedents and consequents, although their interpretability depends on premise design, rule usage, and the operating distribution [7]. Type-2 fuzzy and adaptive neuro-fuzzy sliding-mode controllers have been developed for uncertain surgical-robot tracking [8, 9]; a newer actor–critic sliding-mode approach also provides a Lyapunov analysis for nonlinear systems [10]. Directly trained neuro-fuzzy reinforcement-learning controllers address mobile-robot and other control tasks [11, 12], while a compact neuro-fuzzy method addresses measurement noise [13]. These studies motivate interpretable control but do not establish how faithfully a compressed deep Q-network (DQN) reproduces decisions across sparsely visited states or how that fidelity affects tracking.

The distillation of DQN policies into neuro-fuzzy models is not itself novel. Gevaert et al. [14] demonstrated this approach on benchmarks and compared learning with direct fuzzy control, while Xu et al. [15] extracted interpretable fuzzy rules and evaluated temporal fidelity. Recent concept-based distillation methods have produced sparse surrogate models for image-based policies, whereas temporal distillation methods have compressed both policy representations and their action timing [16]. Those studies target different settings and measures. For uncertain physical tracking, aggregate agreement alone cannot reveal where policy compression fails, whether disagreement affects closed-loop error, or whether teacher guidance changes the rule base relative to direct training. Furthermore, the existing studies do not eliminate the need to distinguish empirical tracking from a formal stability guarantee.

This study contributes the following evidence, without claiming priority for the distillation algorithm itself:

(i) A fixed-premise, head-free Takagi–Sugeno–Kang (TSK) Q-controller whose rule contributions can be inspected, together with an explicit analysis of how action-value errors can change a discrete decision.

(ii) A held-out, common-trace evaluation across nominal, robust, and stress regimes that combines spatial and visitation-conditioned imitation measures with paired closed-loop tracking outcomes.

(iii) An architecture-matched, reward-only comparator with seven membership function (7MF) per input and a post-hoc pruning analysis that test whether teacher supervision changes accuracy and preservation of the learned fuzzy decisions. The 130 reward-training episodes are matched; offline labels, interactions, and computation are not.

The objective of this study is to measure the utility and limitations of interpretable compression rather than to claim the best tracking controller.

2 Methodology

2.1 End-to-End Control and Learning Workflow

Figure 1 distinguishes the physical system from the learning roles. The mass–spring–damper is the controlled plant, while the teacher refers to the DQN policy and the student to its TSK Q-controller approximation, rather than to two different plants. Both policies use the same clipped, noisy observation construction and select a force from the same discrete set. A domain-randomized simulator first trains the teacher, whose visited states provide action, distribution, and Q-value targets for offline student distillation. This is followed by reward-based fine-tuning of the student. An identical TSK architecture is trained directly using reward signals as a comparator. Frozen policies are then evaluated on matched held-out scenarios; teacher-visited states support fidelity analysis, while separate closed-loop rollouts support tracking and pruning analyses.

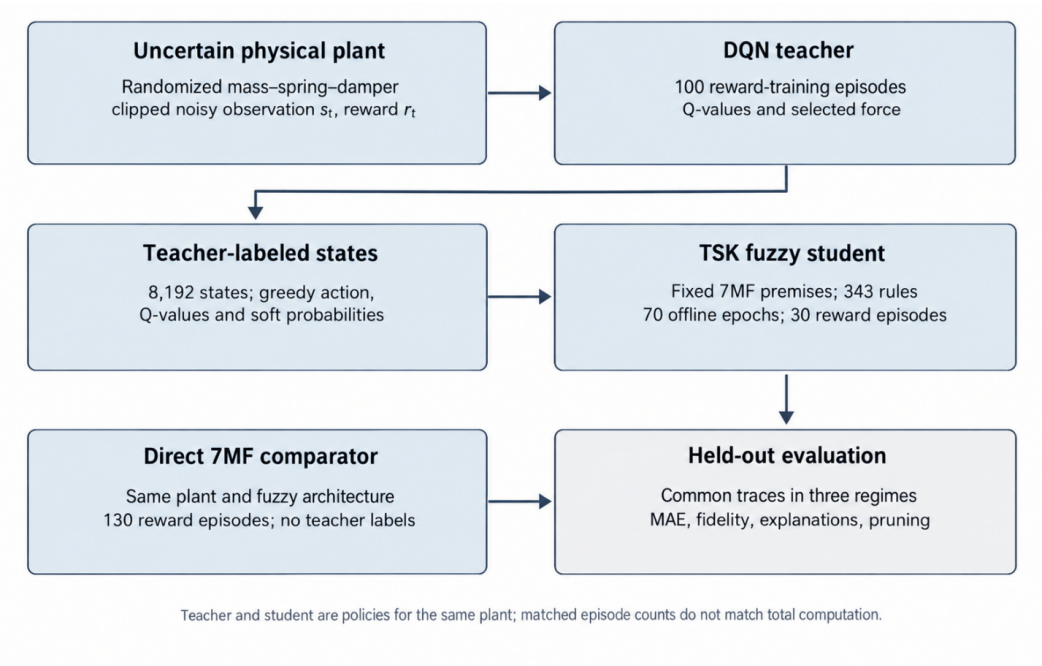


Figure 1. End-to-end methodology

Note: DQN = deep Q-network; TSK = Takagi–Sugeno–Kang; 7MF = seven membership function; MAE = mean absolute error.

The mass–spring–damper is the physical plant; teacher and student are policies for that same plant. The teacher is trained for 100 reward episodes and provides 8,192 labeled states for 70 offline student epochs. The student is subsequently fine-tuned for 30 reward episodes. The direct 7MF comparator receives 130 reward episodes without

teacher targets. Common-trace held-out evaluation distinguishes fidelity on teacher-visited states from closed-loop tracking, rule explanations, and pruning. Episode counts are matched in terms of reward training rather than total interactions or computation.

2.2 Control Objective and Uncertain Plant

The controller tracks a constant set-point r during each episode. The sign of r is sampled with equal probability and its magnitude is drawn from the regime-dependent interval below. The policy observes tracking quantities and chooses a force u_t acting on the plant to reduce $e_t = r - x_t$.

Let x_t, v_t, and d_t denote displacement, velocity, and an external force disturbance, respectively. The parameter Δt is set to 0.02 s. The simulated plant is described by:

$$\begin{aligned} x_{t+1} &= x_t + \Delta t v_t, \\ v_{t+1} &= v_t + \Delta t \frac{u_t + d_t - b v_t - k x_t - k_3 x_t^3 - F_c \tanh(v_t/v_s)}{m} \end{aligned} \tag{1}$$

where, m is the mass, b denotes the viscous damping, k is the linear stiffness, k_3 denotes the cubic stiffness, F_c is the smooth Coulomb-friction scale, and v_s = 0.02 m/s represents the velocity smoothing constant. An episode lasts for a maximum of T = 200 steps and is marked as failed if $|x_t| > 4$ m.

At time step t, the noise-free, error-based tracking-feature vector $z_t \in \mathbb{R}^3$ is defined as:

$$\begin{aligned} z_t &= \begin{bmatrix} e_t \\ \tilde{e}_t \\ I_t \end{bmatrix}, \quad e_t = r - x_t, \quad \tilde{e}_t = -v_t, \\ I_0 &= 0, \quad I_t = \mathrm{clip}\left(I_{t-1} + e_t \Delta t, -20, 20\right), \quad t \geq 1 \end{aligned} \tag{2}$$

where, $z_t \in \mathbb{R}^3$ is a deterministic, physically interpretable policy-feature vector rather than an additional mechanical state or a learned latent variable. Its entries are the instantaneous position error $e_t = r - x_t$ (m), the velocity-based error-rate channel $\tilde{e}_t = -v_t$ (m/s), and the saturated accumulated error I_t (m · s). Because r is constant within an episode, $\dot{e}_t = -v_t$, which explains the second component. The recursion for I_t matches the implementation and limits integral windup; $\mathrm{clip}(q, l, u) = \min\{\max(q, l), u\}$.

Substituting $x_t = r - e_t$ and $v_t = -\tilde{e}_t$ into Eq. (1) gives the deterministic tracking-error dynamics before measurement noise:

$$\begin{aligned} e_{t+1} &= e_t + \Delta t \tilde{e}_t, \\ \tilde{e}_{t+1} &= \tilde{e}_t - \tfrac{\Delta t}{m}\left[u_t + d_t + b\tilde{e}_t - k\left(r - e_t\right) - k_3\left(r - e_t\right)^3 + F_c \tanh\left(\tilde{e}_t/v_s\right)\right] \end{aligned} \tag{3}$$

Thus, the policy's first two features encode the plant's position-error and velocity-error evolution; the integral feature stores clipped error history. This transformation describes the simulator, not a proof that the learned feedback stabilizes it.

Eq. (4) adds the measurement perturbation v_t to this noise-free vector and clips each channel to produce the common teacher and student input s_t. The policy feature $\tilde{e}_t$ differs from the finite-difference quantity Δe_t used only in the reward in Eq. (6).

Independent zero-mean Gaussian perturbations are added component-wise, with the same listed numerical standard deviation in the corresponding coordinate units, and the result is clipped:

$$s_t = \mathrm{clip}\left(z_t + v_t, [-5, -20, -20]^\top, [5, -20, -20]^\top\right), \quad v_{t,i} \sim \mathrm{N}\left(0, \sigma_v^2\right) \tag{4}$$

where, $v_t \in \mathbb{R}^3$ is the measurement noise vector, and σ_v is its per-component standard deviation. The fuzzy antecedent centers remain fixed on the narrower linguistic domain $[-2, 2] \times [-5, 5] \times [-5, 5]$, distinct from the hard policy-input clipping bounds. None of the teacher or distilled 7MF controller observations in the reported evaluation trajectories exceed these premise limits.

The discrete force set and teacher reward are as follows:

$$U = \{-25, -20, -15, -10, -5, 0, 5, 10, 15, 20, 25\} \tag{5}$$

$$\rho_t = -10e_t^2 - 0.1\Delta e_t^2 - 0.005u_t^2 + 0.1, \ \Delta e_t = (e_t - e_{t-1})/\Delta t \tag{6}$$

where, ρ_t is the per-step teacher reward. The nominal, robust, and stress scenarios represent distinct held-out evaluation distributions rather than data folds. Their distinct plant-parameter, reference, noise, and disturbance ranges are specified in Table 1.

Table 1. Training and held-out parameter ranges

Quantity	Training	Nominal	Robust	Stress
Mass, m (kg)	[0.80, 1.20]	[0.95, 1.05]	[0.70, 1.35]	[0.55, 1.60]
Viscous damping, b (N·s/m)	[1.40, 2.60]	[1.90, 2.10]	[1.00, 3.00]	[0.50, 3.80]
Linear stiffness, k (N/m)	[7.50, 12.50]	[9.50, 10.50]	[6.00, 14.00]	[4.00, 18.00]
Cubic stiffness, k_3 (N/m^3)	[0.00, 8.00]	[0.00, 1.50]	[0.00, 10.00]	[5.00, 25.00]
Coulomb-friction scale, F_c (N)	[0.00, 1.00]	[0.00, 0.15]	[0.00, 1.20]	[0.50, 2.00]
Measurement-noise standard deviation, σ_v (m)	0.005	0.00	0.01	0.02
Disturbance standard deviation, σ_d (N)	0.30	0.00	0.50	1.00
Reference magnitude, $\lvert r\rvert$ (m)	[0.30, 0.80]	[0.30, 0.80]	[0.30, 0.80]	[0.50, 1.20]
Initial displacement, x_0 (m)	$[-0.30, 0.30]$	$[-0.25, 0.25]$	$[-0.25, 0.25]$	$[-0.40, 0.40]$

2.3 Deep Q-Network Teacher

The teacher is a rectified linear unit-based Q-network with hidden widths 128, 128, and 64. It is trained for 100 episodes using a replay buffer of 50,000 transitions, Adam with a learning rate of 5×10^{-4}, a discount factor of γ = 0.99, gradient clipping at a norm of 10, and a target-network update every 200 optimizer steps. The double-DQN target is defined as:

$$y_t = \rho_t + \gamma(1-\delta_t)Q_T\left(s_{t+1}, \arg\max_{a'\in\mathcal{U}} Q_T(s_{t+1}, a'; \theta); \theta^-\right) \tag{7}$$

where, y_t is the double-DQN target; $Q_T(\cdot,\cdot;\theta)$ is the teacher action-value function with online parameters θ; $a' \in \mathcal{U}$ is a candidate next action; $\delta_t \in \{0, 1\}$ marks termination; and θ^- denotes target network parameters. The Huber temporal-difference objective $\mathrm{L_{TD}}$ is as follows:

$$\mathrm{L_{TD}} = \mathbb{E}\big[\mathrm{Huber}\big(Q_T(s_t, a_t; \theta) - y_t\big)\big] \tag{8}$$

where, $\mathbb{E}[\cdot]$ denotes the expectation over transitions $(s_t, a_t, \rho_t, s_{t+1}, \delta_t)$ sampled uniformly from the replay buffer. In practice, it is the mini-batch average. $a_t \in \mathcal{U}$ is the force action applied at step t and stored with that transition. The principal teacher is trained with the ranges in Table 1; a nominally trained teacher is used only for the no domain randomization ablation.

2.4 Interpretable Takagi–Sugeno–Kang Q-Controller

For input $i \in \{1, 2, 3\}$ and membership index $j \in \{1, \ldots, M\}$, the fixed Gaussian antecedent is:

$$\mu_{ij}(s_i) = \exp\left[-\tfrac{1}{2}\left(\tfrac{s_i - c_{ij}}{\sigma_{ij}}\right)^2\right]. \tag{9}$$

where, $\mu_{ij}(s_i) \in (0, 1]$ is the membership degree of the i-th policy input s_i in the j-th Gaussian fuzzy set; and c_{ij} and σ_{ij} are the corresponding center and width, respectively. The centers c_{ij} are evenly spaced over $[-2, 2]$, $[-5, 5]$, and $[-5, 5]$ for the three observation coordinates; each width σ_{ij} is 0.9 times the adjacent-center spacing in its coordinate. These premise parameters are fixed after initialization. The small normalizer $\varepsilon = 10^{-8}$ prevents division by zero. Each rule r corresponds to one three-input antecedent-index tuple κ_r. This symbol is distinct from the tracking-feature vector z_t in Eq. (2). The rule's firing and normalized firing are defined as:

$$\phi_r(s) = \textstyle\prod_{i=1}^{3} \mu_{i,\kappa_{ri}}(s_i),\ \alpha_r(s) = \frac{\phi_r(s)}{\sum_{l=1}^{M^3}\phi_l(s)+\varepsilon} \tag{10}$$

The student action values are as follows:

$$Q_S(s, a; \psi) = \textstyle\sum_{r=1}^{M^3} \alpha_r(s) w_{r,a} \tag{11}$$

where, $W = [w_{r,a}]$ has M^3 rows and $|\mathcal{U}| = 11$ columns: row r stores the eleven action values associated with rule r. In addition, ψ denotes the trainable parameters of the student TSK Q-controller. Because the premise parameters are fixed, ψ consists only of the consequent matrix $W = [\{w_{r,a}\}] \in \mathbb{R}^{M^3 \times |\mathcal{U}|}$. The controller with 7MFs per input

therefore has 7^3 = 343 candidate antecedent rules and a 343 × 11 consequent matrix. There is no hidden layer after fuzzy aggregation; therefore, the membership activations and consequents account for the deployed decision directly (Figure 2).

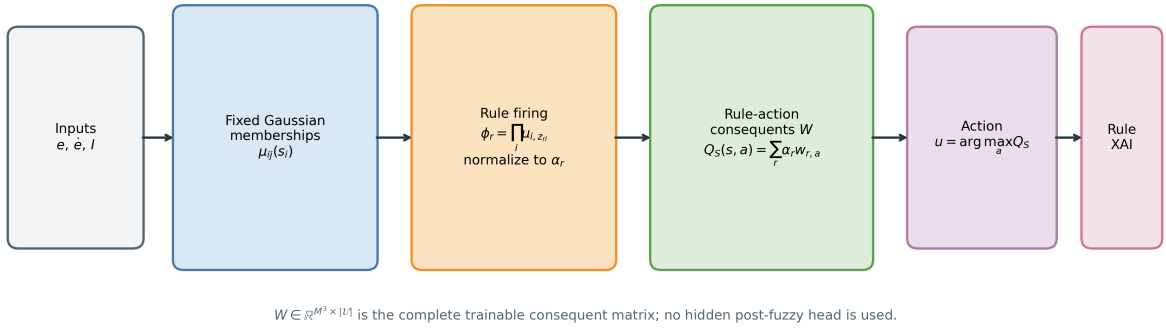


Figure 2. TSK Q-controller shared by the distilled and directly trained 7MF conditions

Note: TSK = Takagi–Sugeno–Kang; 7MF = seven membership function; XAI = Explainable Artificial Intelligence.

2.5 Analytical Properties and Decision Margins

With fixed premises, Eq. (11) is linear in each consequent: $\partial Q_S(s,a)/\partial w_{r,a} = \alpha_r(s)$. Thus, the contribution of a rule to an action value is its normalized firing times its action consequent; inspecting only the largest rule can omit contributions from other active rules. An action decision also depends on the gap between the two largest values, not only on average Q-error. The teacher's greedy action $a_T(s)$, its action margin $g_T(s)$, and the maximum teacher–student action-value difference $d_{TS}(s)$ are defined by:

$$\begin{aligned} g_T(s) &= Q_T(s,a_T) - \max_{a\in\mathcal{U},\, a\neq a_T} Q_T(s,a), \\ d_{TS}(s) &= \max_{a\in\mathcal{U}} Q_T(s,a) - Q_S(s,a), \\ g_T(s) > 2d_{TS}(s) &\Rightarrow \quad \arg\max_{a\in\mathcal{U}} Q_S(s,a) = a_T(s) \end{aligned} \tag{12}$$

The implication follows because an error of at most d_{TS} can decrease the teacher's top value and increase a competing value by at most d_{TS} each. It explains why small value errors near a discrete switching boundary can change the selected force, but it is a statewise sufficient condition, not a stability or safety guarantee. The same condition applies when comparing an unpruned fuzzy policy with its pruned version, using their own maximum Q-difference and the unpruned margin. The same controller architecture (Figure 2) is used for both distilled and direct training, with differences limited to the learning signals and training schedules.

2.6 Distillation and Architecture-Matched Direct Training

The distillation set contains 8,192 states. At each collection step, the greedy teacher action is selected with a probability of 0.8 and a random exploratory action with a probability of 0.2. At a temperature of τ = 2, the teacher provides the following:

$$a_T(s) = \arg\max_a Q_T(s,a), \quad p_T^\tau(a \mid s) = \frac{\exp\left(Q_T(s,a)/\tau\right)}{\sum_{a'} \exp\left(Q_T(s,a')/\tau\right)} \tag{13}$$

The student is optimized for 70 offline epochs using the following:

$$L_{\text{distill}} = L_{\text{CE}} + 0.25\tau^2 D_{\text{KL}}\left(p_T^\tau \| p_S^\tau\right) + 0.05\ \text{Huber}(Q_S, Q_T) + L_{\text{aux}} \tag{14}$$

Subsequently, 30 episodes of direct DQN fine-tuning are performed. In this formulation, L_{CE} is cross-entropy between the teacher's greedy action and the student's untampered action logits; the Kullback–Leibler (KL) term compares the two temperature-τ action distributions, and the Huber term matches their raw Q-values. The auxiliary term is defined as:

$$L_{\text{aux}} = \frac{\lambda_1}{M^3|U|}\|W\|_1 + \lambda_m L_{\text{mono}}, \quad \lambda_1 = 10^{-6}, \quad \lambda_m = 10^{-3} \tag{15}$$

where, $\|W\|_1/(M^3|U|)$ is the mean absolute consequent magnitude. For a state s, let $\bar{u}(s) = \sum_{a\in U} a\, p_S^2(a \mid s)$ be the soft expected force. The monotonic penalty is the mean over training states and the first two observation

coordinates of $\max\{0, \bar{u}(s - 0.05e_i) - \bar{u}(s + 0.05e_i)\}$, where e_i is the i-th coordinate vector. It discourages a decreasing soft force when the error or error-rate input increases; it does not impose a hard monotonicity constraint. Premise parameters remain fixed.

The architecture-matched direct baseline uses the same premise grid with 7 MFs, W dimensions, force set, DQN loss, domain randomization, online DQN optimizer settings, seeds, and auxiliary coefficients, but receives no teacher actions, distributions, or Q-values. It is trained using task rewards for 130 episodes, matching the combined 100 episodes of teacher training and 30 episodes of student fine-tuning in the distillation schedule. The distilled condition additionally collects 8,192 teacher-labeled simulator states and optimizes them for 70 offline epochs. Architecture and reward-training episode count are therefore matched, whereas total simulator interactions, gradient updates, computation, and access to supervision are not. The same direct model is used for the direct-policy pruning analysis.

2.7 Fidelity, Explanations, and Pruning

Teacher–student fidelity is evaluated on policy-input states visited by the matched reference teacher under held-out scenarios:

$$A_{TS} = \frac{1}{|D_T|} \sum_{s \in D_T} \mathbb{1}\left\{ \arg\max_{a \in U} Q_S(s,a) = \arg\max_{a \in U} Q_T(s,a) \right\},$$
$$K_{TS} = \frac{1}{|D_T|} \sum_{s \in D_T} D_{\mathrm{KL}}\big(p_T^\tau(\cdot \mid s) \| p_S^\tau(\cdot \mid s)\big) \tag{16}$$

where, the subscripts T and S denote the teacher and student, respectively. Within each seed and evaluation regime, the multiset D_T contains the clipped policy-input vectors s encountered immediately before the matched DQN teacher selects an action on the held-out scenarios, and $|D_T|$ = 16×200 = 3200 is its sample count. The functions $Q_T(s,a)$ and $Q_S(s,a)$ are the teacher and student action values for action $a \in U$; $\arg\max_{a \in U}$ selects the greedy force; and $\mathbf{1}\{\cdot\}$ equals one when the two greedy actions match and zero otherwise. For $X \in \{T, S\}$, $p_X^\tau(a \mid s) = \exp\big(Q_X(s,a)/\tau\big) / \sum_{a' \in U} \exp\big(Q_X(s,a')/\tau\big)$ is the softened categorical distribution over the complete action set at temperature τ = 2, where exp denotes the exponential function. The dot in $p_X^\tau(\cdot \mid s)$ denotes that complete eleven-action probability vector, and $D_{\mathrm{KL}}(p\|q) = \sum_{a \in U} p(a) \ln\big[p(a)/q(a)\big]$ is the ordered KL divergence using the natural logarithm and expressed in nats. Consequently, $A_{TS} \in [0, 1]$ is the empirical greedy-action agreement, with 1 denoting perfect action imitation, whereas $K_{TS} \geq 0$ is the mean distributional mismatch, with 0 denoting identical softened action preferences. All matched teacher rollouts completed 200 steps; therefore, no padding or imputation was required. The metrics are computed separately for each seed; the reported means and 95% confidence intervals (CIs) based on Student's t-distribution are calculated from ten seed-level values, representing 32,000 states in total per regime. Eq. (16) therefore measures imitation over teacher-visited states separately from closed-loop task performance. It does not assert fidelity over unvisited portions of the clipped input box or over states visited only by the student. The two measures are complementary: A_{TS} tests the selected action but discards non-maximal values, while K_{TS} retains the teacher's relative preferences over all eleven actions and can expose discrepancies even when the greedy actions agree. Pearson correlation between the complete teacher and student Q-vectors supplies a third aggregate view, and the subsequent maps and paired rollouts localize mismatch and assess its observational association with closed-loop mean absolute error (MAE).

Two analyses are conducted to localize failures. First, states are projected onto the error–error-rate plane and grouped into bins to compare teacher visitation counts with action-disagreement rates; bins with fewer than 12 states are omitted from the rate map. Second, the full three-dimensional clipped policy space is divided into 20 × 20 × 20 occupancy bins. Every teacher-visited state inherits the count of its bin and is assigned to a visitation-density quartile, ranging from Q1 (rarest) to Q4 (most visited). The disagreement is computed separately within each seed and the mean with a 95% CI based on Student's t-distribution is reported using ten seed-level values.

To assess whether fidelity is associated with control, the teacher and student are also rolled out separately on the same scenario and exogenous traces. For each of the 160 scenarios per regime, the disagreement along the teacher trajectory is related to $\Delta\mathrm{MAE} = \mathrm{MAE}_S - \mathrm{MAE}_T$. The pooled Spearman correlation, mean seed-wise Spearman correlation with 95% CIs based on Student's t-distribution across ten within-seed correlations, and the number of scenarios meeting or exceeding the 75th-percentile disagreement threshold (including ties) for which $|\Delta\mathrm{MAE}| \leq 0.02$ are reported. This analysis is observational and is not interpreted causally.

Intrinsic rule summaries use mean firing, the count of rules whose trajectory-mean firing exceeds 0.01, the mean number of rules with instantaneous firing above 0.05, and the fraction of rules ever exceeding 0.05. For pruning, rules are ranked from full-policy trajectories using mean firing, maximum firing, and active fraction. Only the top $K \in \{8, 16, 32, 64, 96, 128\}$ rules are retained, remaining firings are renormalized, and both closed-loop MAE and action fidelity to the corresponding full fuzzy policy are evaluated. The reported pruning fidelity is pooled across all three regimes for each seed; it is not a regime-specific quantity. Rule ranking and pruned-policy evaluation use the

same held-out scenario collection. Therefore, pruning is presented as a post-hoc operating-envelope analysis rather than independent validation on a second scenario set.

2.8 Experimental Protocol

All final experiments use seeds 0–9. Each seed contains 16 independently sampled scenarios in each evaluation regime, giving 160 rollouts per controller and regime. For a given seed–regime–scenario tuple, every controller receives the same initial condition, plant parameters, reference, disturbance trace, and observation-noise trace. This pairing prevents exogenous variation from being mistaken for a controller effect. Evaluation regimes are held out from training and use the ranges in Table 1.

Baselines are a hand-crafted two-input fuzzy controller, a proportional–integral–derivative (PID) controller (K_p = 35, K_d = 8, K_i = 2), a nominal discrete-time linear quadratic regulator (LQR), the domain-randomized DQN teacher, distilled five-membership-function and 7MF students, a 7MF student trained without domain randomization, and a directly trained 7MF model. The PID and LQR controllers are deliberately strong for this mechanical tracking problem; they prevent improvements in interpretability from being presented as optimal control performance.

The primary metric is mean absolute tracking error:

$$\mathrm{MAE} = \frac{1}{T}\sum_{t=1}^{T}|e_t| \tag{17}$$

Secondary metrics include root mean square error, integrated absolute error, the discrete-time integral of squared force (control energy), settled fraction, and failure rate. CIs are computed from the ten seed-level means using the critical value of Student's t-distribution with nine degrees of freedom. Paired comparisons first average scenario differences within each seed and then apply a two-sided Wilcoxon signed-rank test over ten seed differences. The three direct-versus-distilled regime tests are corrected using the Holm method. Scenario-level win rates and pooled Spearman tests are descriptive complements; the inferential unit for the main performance comparison remains the seed.

3 Results

3.1 Teacher–Student Fidelity and Its Limitations

The 7MF student provides a useful but incomplete approximation of its teacher. Action agreement is 59.79% $\pm$ 7.44 percentage points in the nominal regime, 53.31% $\pm$ 11.30 in the robust regime, and 29.79% $\pm$ 7.80 in the stress regime (mean $\pm$ 95% CI half-width over ten seeds). The softened KL divergences are 0.0628 $\pm$ 0.0111, 0.0638 $\pm$ 0.0094, and 0.1821 $\pm$ 0.0140, respectively, while the corresponding Q-vector Pearson correlation coefficients are 0.2168 $\pm$ 0.0291, 0.2248 $\pm$ 0.0332, and 0.1009 $\pm$ 0.0148. Thus, the stress regime changes both the teacher's visited region and the student's approximation quality. The student should not be described as a lossless clone.

Figure 3 shows the spatial structure hidden by the averages. Nominal and robust teacher trajectories concentrate along a narrow error–error-rate manifold, while stress trajectories spread farther from it. Disagreement is not uniformly distributed: it is prominent along parts of the transition band where small changes in action values change the discrete argmax, and it expands in the shifted stress occupancy. The integral input is included in the fidelity calculation but marginalized in this visualization.

The full three-dimensional density analysis supports, but does not oversimplify, the same pattern (Figure 4a). From Q1 to Q4, disagreement rates are 51.4%, 49.1%, 26.6%, and 34.9% under nominal conditions; 55.9%, 53.7%, 44.4%, and 37.7% under robust conditions; and 77.1%, 70.5%, 73.0%, and 62.5% under stress conditions. Rare states are generally harder, especially in robust and stress regimes, but the nominal sequence is not monotonic. Density is therefore a descriptive correlate rather than a complete explanation; unmeasured factors such as local action-value margins and the marginalized integral input may also contribute.

Disagreement is positively associated with a measurable closed-loop consequence (Figure 4b). The pooled scenario-level Spearman correlation coefficients between teacher-trajectory disagreement and the MAE gap (ΔMAE) are 0.542, 0.635, and 0.649 for the nominal, robust, and stress regimes ($p < 2 \times 10^{-13}$ in each case). More conservatively, the means of the ten within-seed correlations are 0.56, 0.52, and 0.60, with corresponding 95% CIs of [0.31, 0.80], [0.33, 0.71], and [0.45, 0.75], respectively; signed-rank tests against zero yield $p \leq 0.004$. Different action sequences can occasionally yield comparable tracking: among scenarios at or above the 75th-percentile disagreement threshold, including ties, 3/42 nominal, 1/40 robust, and 0/40 stress scenarios satisfy $|\Delta\mathrm{MAE}| \leq 0.02$. Those exceptions occur only in the easier regimes and are consistent with closed-loop recovery or action differences with limited net effect. None of the highest-disagreement stress scenarios satisfies the low-gap threshold. These associations are observational and are not interpreted causally.

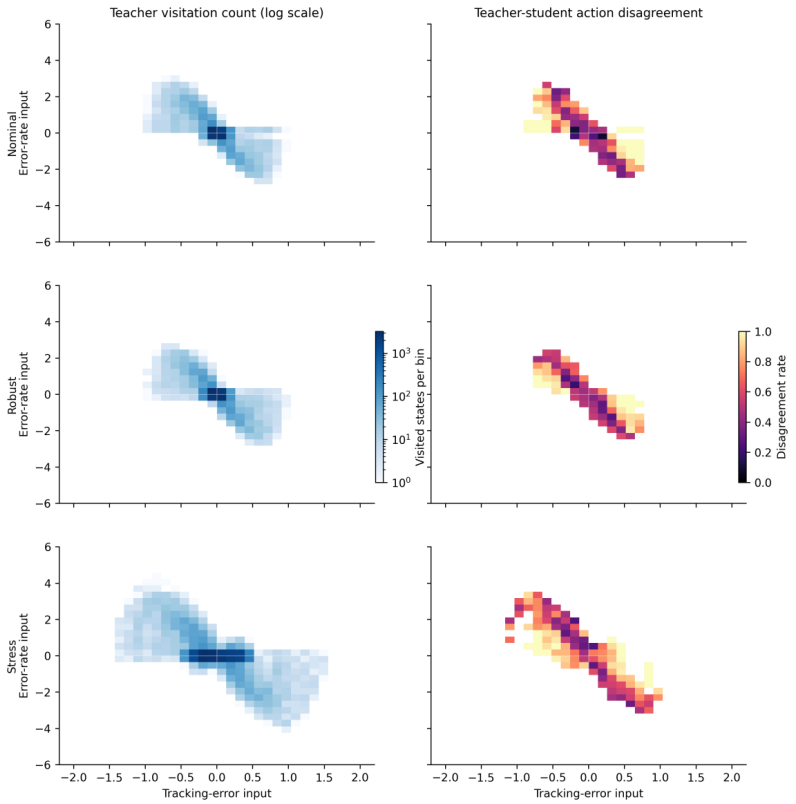
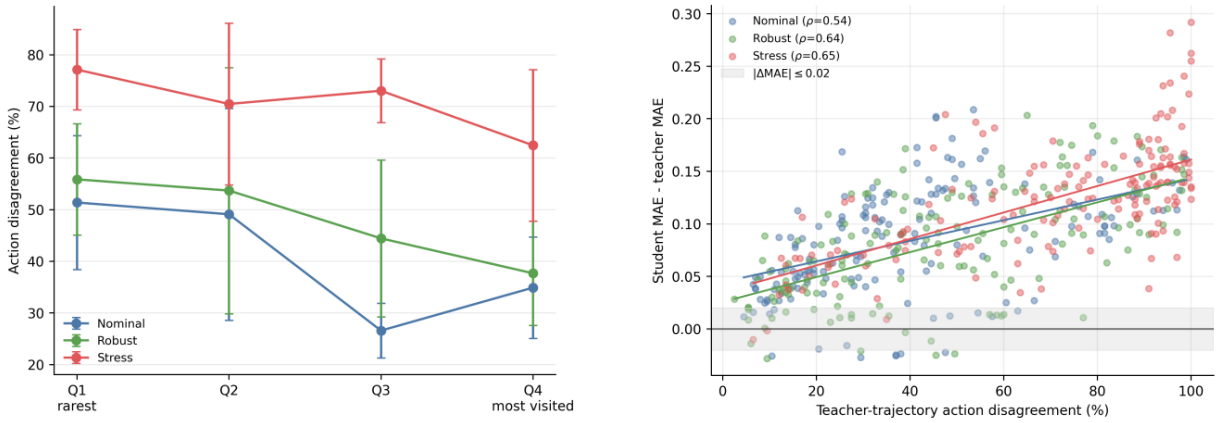


Figure 3. Teacher visitation and student disagreement in the error–error-rate plane

Note: Teacher-visited-state counts are shown on a logarithmic scale (left), and action-disagreement rates are shown for bins containing at least 12 states (right).



(a) (b)

Figure 4. Consequences of teacher–student divergence: (a) Disagreement rates across quartiles of three-dimensional teacher-visitation density, shown as seed means with 95% CIs; (b) Scenario-level disagreement plotted against the student-minus-teacher $\Delta\mathrm{MAE}$ under common traces

Note: Legend values indicate pooled Spearman correlation coefficients, and the shaded band represents $|\Delta\mathrm{MAE}| \leq 0.02$. CIs = confidence intervals; MAE = mean absolute error.

3.2 Closed-Loop Performance

Figure 5 shows the scenario-level MAE distributions. The MAEs for the distilled 7MF controller are 0.1589 (95% CI half-width: 0.0178), 0.1552 (0.0175), and 0.3437 (0.0284) across the three regimes. The hand-crafted fuzzy controller obtains 0.2572, 0.2558, and 0.4463. Therefore, the distilled 7MF controller reduces the MAEs by 38.2%, 39.4%, and 23.0%. None of the unpruned controllers in Figure 5 produces a displacement failure.

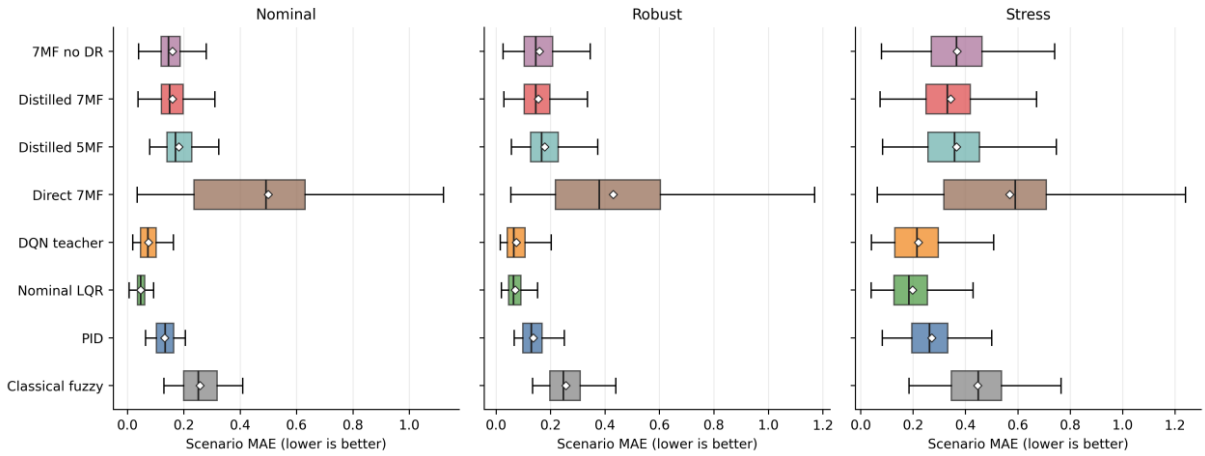


Figure 5. Closed-loop distributions over ten seeds and 160 held-out scenarios per regime

Note: Diamonds indicate pooled means, with lower values representing better tracking performance. The directly trained and distilled 7MF controllers share the architecture shown in Figure 2. 7MF = seven membership function; DR = domain randomization; 5MF = five membership function; DQN = deep Q-network; LQR = linear quadratic regulator; PID = proportional–integral–derivative; MAE = mean absolute error.

The interpretability cost remains visible. The PID controller achieves MAEs of 0.1317, 0.1363, and 0.2710; the nominal LQR achieves 0.0471, 0.0702, and 0.1977; and the DQN teacher achieves 0.0748, 0.0742, and 0.2200. Accordingly, the proposed student is not the most accurate controller for this plant. Its value is the combination of improved fuzzy control, explicit rules, and measurable compression. Mean control energy values are 107.5, 114.5, and 274.0 for the distilled 7MF policy, compared with 850.1, 830.7, and 989.4 for the directly trained 7MF policy; under the tested schedules, distillation improves tracking and reduces across-seed outcome variability rather than merely trading error for more actuation.

3.3 Effects of Distillation

The direct 7MF controller yields MAEs of 0.4989 $\pm$ 0.1971, 0.4299 $\pm$ 0.1296, and 0.5690 $\pm$ 0.1383 (seed mean $\pm$ 95% CI half-width). The paired seed-level differences between the distilled and directly trained controllers are -0.3400 with 95% CI [-0.5344, -0.1456], -0.2748 [-0.3941, -0.1554], and -0.2253 [-0.3486, -0.1021]. Distillation outperforms direct training in 86.3%, 89.4%, and 71.3% of paired scenarios; all three seed-level signed-rank comparisons remain significant after Holm correction ($p_{\mathrm{adj}} \leq 0.006$). The wide direct-training intervals in Figure 6 show greater across-seed performance variability for direct training.

Capacity and randomization ablations qualify this result. The distilled five-membership-function student achieves MAEs of 0.1818, 0.1780, and 0.3652. Therefore, the larger fixed vocabulary improves task accuracy. However, the five-membership-function student has higher teacher action agreement in nominal and robust regimes (65.2% and 60.4%) than the 7MF student (59.8% and 53.3%). This reversal shows that aggregate action agreement and closed-loop performance are distinct and is consistent with task-relevant corrections reducing exact imitation; the mechanism was not isolated through an experiment without fine-tuning. Removing training-domain randomization changes has little effect on nominal and robust MAEs (0.1593 and 0.1587) but increases stress MAE to 0.3673. The unadjusted stress comparison is $p = 0.037$ and is treated as exploratory rather than confirmatory after considering the family of ablations.

The lower panels of Figure 6 compare pruning within the same fuzzy architecture. For the distilled 7MF controller, pooled action fidelity to the full policy is 90.5% $\pm$ 1.87 percentage points with 64 rules, 95.9% $\pm$ 0.78 with 96 rules, and 97.6% $\pm$ 0.89 with 128 rules. At 64 rules, the MAEs are 0.1679, 0.1619, and 0.3520, compared with 0.1589, 0.1552, and 0.3437 for the full policy. The 96- and 128-rule configurations nearly recover the full-policy performance, with overlapping CIs; no formal equivalence or paired K-versus-full hypothesis test is claimed.

Direct training produces a markedly less prune-tolerant policy under this operating-envelope analysis. Its pooled fidelity is only 42.8% at 64 rules, 57.2% at 96 rules, and 66.4% at 128 rules. Even the 128-rule direct policy has MAEs of 0.6622, 0.5651, and 0.7397, exceeding those of the variable full direct controller, and it produces two displacement failures among 160 stress rollouts. The difference suggests that teacher-supervised training may influence which rules govern frequently encountered decisions; its cause cannot be isolated from the additional offline optimization in this design.

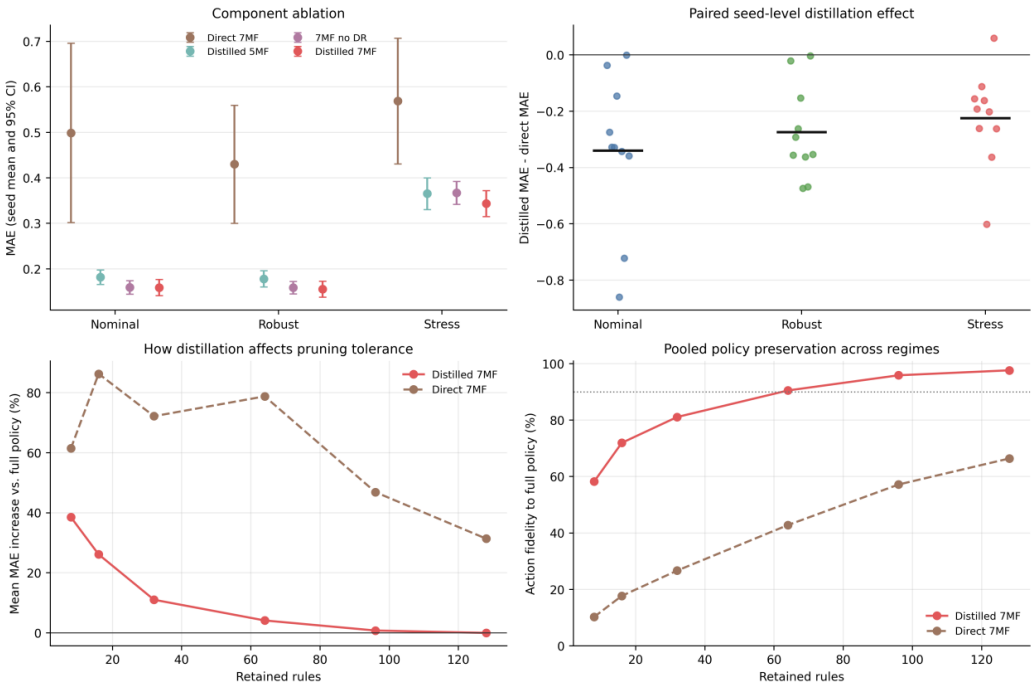


Figure 6. Ablation and pruning analysis

Note: The upper-left panel compares membership-function capacity, domain randomization, and direct training. The upper-right panel shows paired seed-level differences in MAE between the distilled and directly trained controllers across ten seeds. The lower panels show changes in MAE and action fidelity following rule pruning. Fidelity is pooled across evaluation regimes within each seed. MAE = mean absolute error; CI = confidence interval; 7MF = seven membership function; 5MF = five membership function; DR = domain randomization.

3.4 Rule-Level Interpretability

The full 7MF grid contains 343 antecedent combinations, but the distilled student uses only 27.2 ± 0.4 effective rules, with approximately 3.15 rules exceeding the instantaneous activation threshold per state and $9.9\% \pm 1.5$ percentage points of the rule base covered. The directly trained 7MF controller uses approximately 30.1 effective rules, 3.44 active rules per state, and 23.8% coverage. Similar effective counts therefore do not imply similar rule allocation or pruning tolerance (Figure 7).

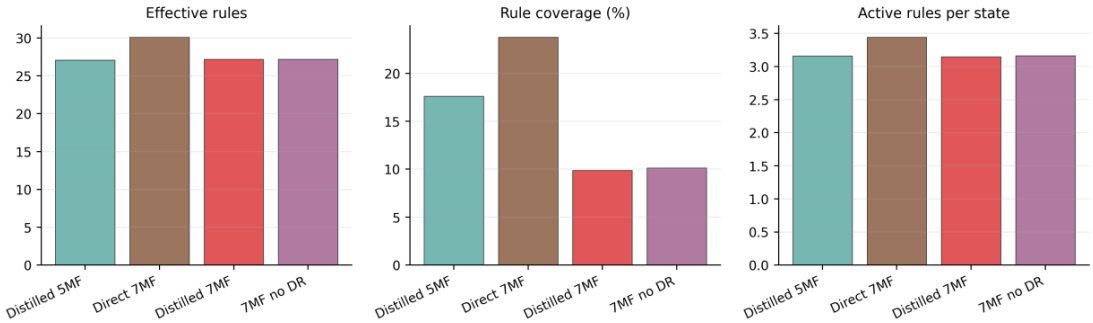


Figure 7. Rule-usage summaries

Note: Distilled models remain sparse despite different candidate capacities, whereas the directly trained 7MF controller visits a broader fraction of the same 343-rule grid. DR = domain randomization; 5MF = five-membership-function controller; 7MF = seven membership function.

Figure 8 illustrates the explanation interface for the final seed-0 distilled 7MF checkpoint. Mean firing ranks globally used rules over the held-out common-trace scenarios, while the trajectory view reports the leading local activations together with the selected force for nominal scenario 0. A single rule should not be read as a complete causal explanation because the final Q-values aggregate all normalized firings. Reporting the leading rules, their strengths, and the final action preserves that distinction.

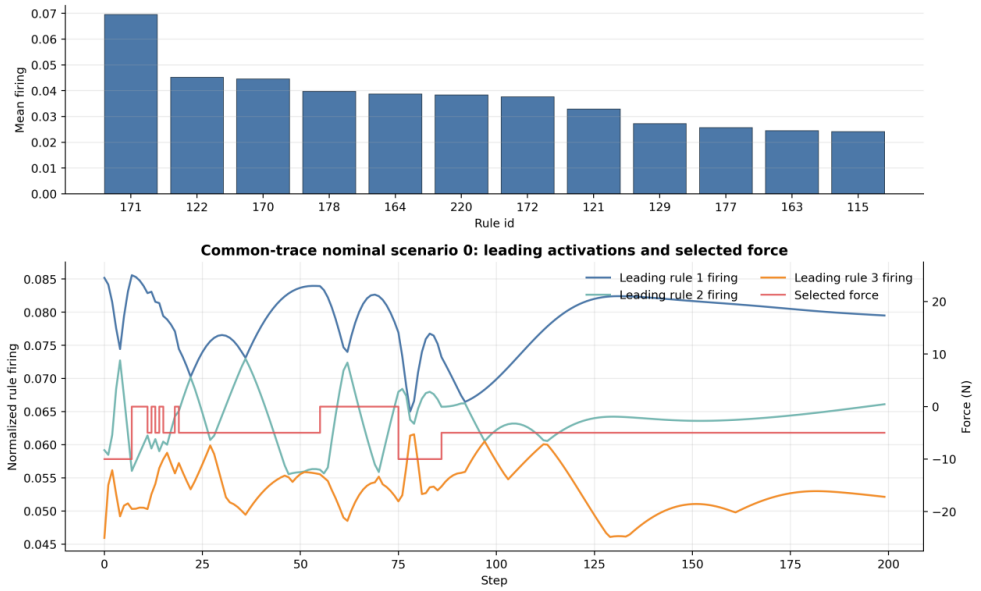


Figure 8. Intrinsic rule-level explanation for the final seed-0 distilled 7MF checkpoint

Note: The upper panel shows the most frequently activated rules across the held-out common-trace scenarios, while the lower panel shows the three leading rule activations and the selected control force for nominal scenario 0. 7MF = seven membership function.

The same final seed-0 policy can be audited globally by fixing the integral input at zero and sweeping error and error rate (Figure 9). The surface reveals switching boundaries and sign structure that are inaccessible from an aggregate score. It is a diagnostic visualization rather than a stability certificate.

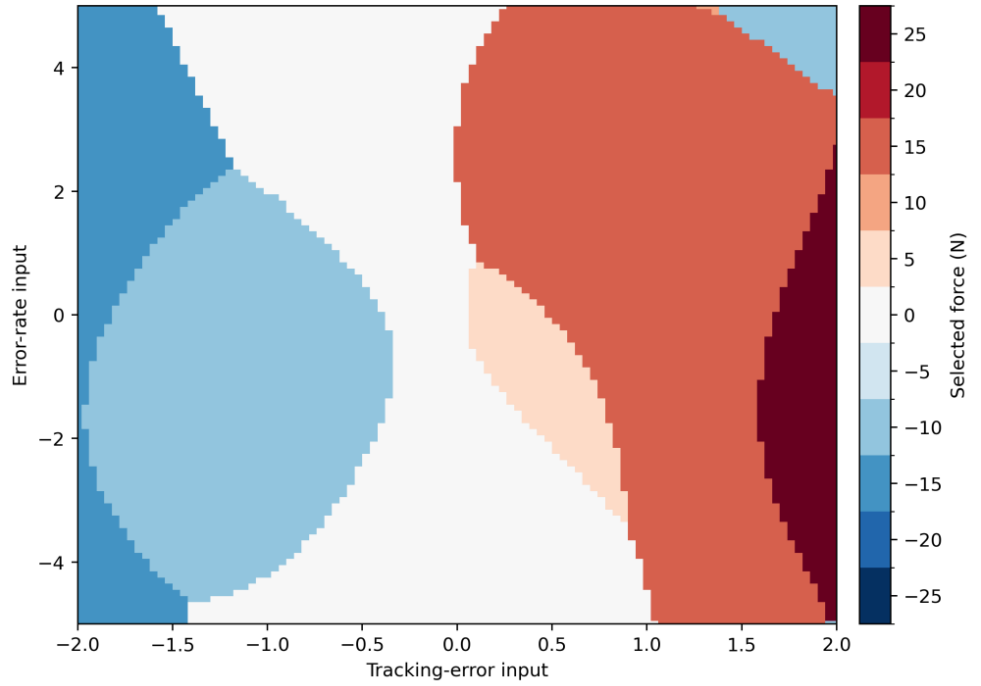


Figure 9. Selected force of the final seed-0 distilled 7MF checkpoint over the error–error-rate plane with integral input fixed at 0

Note: 7MF = seven membership function.

4 Discussion

4.1 Fidelity and Closed-Loop Consequences

The discrete argmax in Eq. (12) makes action agreement sensitive to the gap between the best and second-best action values. The decision-margin condition is sufficient but was not evaluated as a certified margin on every rollout;

the mismatch maps instead provide empirical localization. Action agreement in the stress regime decreases from 59.8% under nominal conditions to 29.8%, while the softened KL divergence increases from 0.0628 to 0.1821. The student–teacher MAE gap also grows from 0.0841 m under nominal conditions to 0.1237 m under stress conditions. These concurrent changes are consistent with the shifted occupancy and stronger uncertainty, but they do not isolate a single cause.

Action agreement is not itself a measure of control performance. The five-membership-function student agrees more often with its teacher than the 7MF student in nominal and robust conditions (65.2% versus 59.8%, and 60.4% versus 53.3%), yet yields higher MAEs (0.1818 versus 0.1589, and 0.1780 versus 0.1552). Conversely, a few high-disagreement scenarios retain nearly the same MAEs because different force sequences can have limited net effect or can be corrected later. The positive associations between disagreement and MAE are observational: difficult scenarios can increase both quantities. The nominal density-quartile pattern is also non-monotonic, indicating that low visitation alone does not explain each mismatch.

4.2 Benefits and Cost of Teacher Guidance

Compared with the directly trained 7MF controller with the same architecture, distillation reduces MAEs by 68.1%, 63.9%, and 39.6% in nominal, robust, and stress conditions. The ten-seed paired differences remain significant after Holm correction. The distilled policy also uses less measured control energy and shows narrower across-seed MAE intervals. However, neither result implies that it is the best controller, as the PID controller, the LQR, and the DQN teacher are more accurate in every regime. The comparison matches the fuzzy architecture and 130 reward-training episodes, not total simulator interactions, optimizer updates, or computation. Thus, the result supports an advantage of the complete teacher-guided schedule tested in this study rather than a claim of superiority under matched computational budgets.

The pruning result provides an additional structural observation. Retaining 64 of 343 rules (18.7%) preserves 90.5% of full-student actions, with MAE increases of 0.0090, 0.0067, and 0.0083 m across the three regimes. A directly trained model retains only 66.4% of its actions with 128 rules and has two stress failures after pruning. Similar full-model effective-rule counts therefore conceal different allocations of decision weight. The action-margin condition explains why rule removal can be harmless where the value gap is large, but the condition was not used to certify these pruned trajectories. The observed MAE changes do not establish equivalence.

4.3 Novelty and Validity Limits

Gevaert et al. [14] investigated the distillation of DQN policies into neuro-fuzzy models on benchmark tasks, while Xu et al. [15] investigated interpretable fuzzy-rule extraction. The present study extends these approaches through an integrated evaluation of uncertain nonlinear tracking, teacher-visited-state fidelity, paired closed-loop outcomes, and architecture-matched pruning. Concept-based and temporal distillation pursue different compression objectives [16]. Unlike control designs with a Lyapunov analysis [10], the present policy has no proved stability, robust invariance, or safety guarantee.

The evidence covers one simulated plant family, one discrete force set, and ten seeds. The fixed premise grid can limit transfer outside its linguistic domain. Teacher-visited-state fidelity omits states reached only by the student; two-dimensional maps marginalize the integral input, and density quartiles depend on binning. Pruning ranks and evaluates rules on the same held-out scenarios, indicating that it is a post-hoc operating-envelope result rather than independent validation. Hardware experiments, training under matched computational budgets, and independent pruning scenarios are needed before generalizing the reported gains.

5 Conclusion

A fixed-premise TSK Q-controller can compress a DQN policy into directly inspectable rule–action values, but the resulting behavior must be assessed in a closed loop. Across nominal, robust, and stress regimes, the distilled 7MF policy achieved MAEs of 0.1589, 0.1552, and 0.3437 m. These values were 68.1%, 63.9%, and 39.6% lower than those of the architecture-matched, directly trained 7MF controller, which achieved MAEs of 0.4989, 0.4299, and 0.5690 m under the tested schedules; paired ten-seed comparisons remained significant after Holm correction ($p_{\mathrm{adj}} \leq 0.006$). Relative to a manual fuzzy controller, the corresponding reductions were 38.2%, 39.4%, and 23.0%. Pruning to 64 of 343 rules preserved 90.5% of full-student actions, although the MAE increased slightly. The distillation was not lossless: teacher-action agreement was only 29.8% under stress conditions, while the PID controller, the LQR, and the DQN teacher remained more accurate. The result is an auditable policy with quantified trade-offs between accuracy and fidelity, rather than a stability guarantee.

Future work should test additional plants and continuous-action policies, compare matched total computation and simulator interactions, and validate rule pruning on independent shifted scenarios. Hardware-in-the-loop studies and safety constraints or formal stability analysis are needed before deployment.

Author Contributions

Conceptualization, M.M. and M.K.; methodology, M.M. and T.B.D.; software, T.B.D. and M.M.; validation, M.M. and M.K.; formal analysis, M.K.; investigation, M.K.; resources, M.K.; data curation, M.M.; writing—original draft preparation, M.M., T.B.D., and M.K.; writing—review and editing, M.K.; visualization, M.K.; supervision, M.K.; project administration, M.K. All authors have read and agreed to the published version of the manuscript.

Data Availability

The code, trained models, scenario-level metrics, common exogenous traces, and analysis outputs supporting the reported findings are available from the corresponding author upon reasonable request.

Conflict of Interest

The authors declare no conflicts of interest.

References

[1] M. Garcia and W. Dong, "Reinforcement learning-based optimal control of uncertain nonlinear systems," *Int. J. Control*, vol. 97, no. 12, pp. 2839–2850, 2024. https://doi.org/10.1080/00207179.2024.2305727

[2] W. A. Suttle, V. K. Sharma, K. C. Kosaraju, S. Seetharaman, J. Liu, V. Gupta, and B. M. Sadler, "Sampling-based safe reinforcement learning for nonlinear dynamical systems," in *Proceedings of the 27th International Conference on Artificial Intelligence and Statistics (AISTATS)*, vol. 238, Valencia, Spain, 2024, pp. 4420–4428. https://proceedings.mlr.press/v238/suttle24a.html

[3] T. Slawik, B. Wehbe, L. Christensen, and F. Kirchner, "Deep reinforcement learning for path-following control of an autonomous surface vehicle using domain randomization," *IFAC-PapersOnLine*, vol. 58, no. 20, pp. 21–26, 2024. https://doi.org/10.1016/j.ifacol.2024.10.027

[4] Y. Jiang, C. Li, W. Dai, J. Zou, and H. Xiong, "Variance reduced domain randomization for reinforcement learning with policy gradient," *IEEE Trans. Pattern Anal. Mach. Intell.*, vol. 46, no. 2, pp. 1031–1048, 2024. https://doi.org/10.1109/TPAMI.2023.3330332

[5] S. Milani, N. Topin, M. Veloso, and F. Fang, "Explainable reinforcement learning: A survey and comparative review," *ACM Comput. Surv.*, vol. 56, no. 7, pp. 1–36, 2024. https://doi.org/10.1145/3616864

[6] C. Glanois, P. Weng, M. Zimmer, D. Li, T. Yang, J. Hao, and W. Liu, "A survey on interpretable reinforcement learning," *Mach. Learn.*, vol. 113, no. 8, pp. 5847–5890, 2024. https://doi.org/10.1007/s10994-024-06543-w

[7] N. Yogeesh, S. I. Mohammad, N. Raja, R. Chetana, P. William, A. Vasudevan, B. A. Qraini, and M. T. Alshurideh, "From crisp to fuzzy: A comparative review of statistical and fuzzy approaches to problem solving," *Appl. Math. Inf. Sci.*, vol. 19, no. 3, pp. 647–658, 2025. http://dx.doi.org/10.18576/amis/190313

[8] S. Sachan and P. Swarnkar, "Intelligent fractional order sliding mode based control for surgical robot manipulator," *Electronics*, vol. 12, no. 3, p. 729, 2023. https://doi.org/10.3390/electronics12030729

[9] S. Sachan, T. Shukla, P. Swarnkar, and A. Adhikari, "Intelligent control of surgical robot for telesurgery: An application to smart healthcare systems," *IET Circuits Devices Syst.*, vol. 2024, no. 1, p. 2119507, 2024. https://doi.org/10.1049/2024/2119507

[10] O. Youssef, M. Wafa, and R. Shalaby, "Reinforcement learning-enhanced adaptive sliding mode control for nonlinear systems," *Complex Intell. Syst.*, vol. 11, no. 8, p. 351, 2025. https://doi.org/10.1007/s40747-025-01978-7

[11] C.-F. Juang and Z.-B. You, "Reinforcement learning of an interpretable fuzzy system through a neural fuzzy actor-critic framework for mobile robot control," *IEEE Trans. Fuzzy Syst.*, vol. 32, no. 6, pp. 3655–3668, 2024. https://doi.org/10.1109/TFUZZ.2024.3380824

[12] T.-L. Nguyen, N. V. Thin, and S. Lee, "Modified triplet-average deep deterministic policy gradient for interpretable neuro-fuzzy deep reinforcement learning," *J. Franklin Inst.*, vol. 362, no. 7, p. 107653, 2025. https://doi.org/10.1016/j.jfranklin.2025.107653

[13] M. M. Ferdaus, A. J. Al-Mahasneh, S. G. Anavatti, and J. Senthilnath, "A compact meta-learned neuro-fuzzy technique for noise-robust nonlinear control," *Appl. Soft Comput.*, vol. 166, p. 112149, 2024. https://doi.org/10.1016/j.asoc.2024.112149

[14] A. Gevaert, J. Peck, and Y. Saeys, "Distilling deep RL models into interpretable neuro-fuzzy systems," in *2022 IEEE International Conference on Fuzzy Systems (FUZZ-IEEE)*, Padua, Italy, 2022, pp. 1–8. https://doi.org/10.1109/FUZZ-IEEE55066.2022.9882791

[15] P. Xu, Z. Deng, J. Wang, Q. Zhang, K. S. Choi, and S. Wang, "Transfer representation learning with TSK fuzzy system," *IEEE Trans. Fuzzy Syst.*, vol. 29, no. 3, pp. 649–663, 2021. https://doi.org/10.1109/TFUZZ.2019.2958299

[16] M. Farhadi and Y. Yang, “TKD: Temporal knowledge distillation for active perception,” in *2020 IEEE Winter Conference on Applications of Computer Vision (WACV 2020)*, Snowmass, USA, 2020, pp. 942–951. https://doi.org/10.1109/WACV45572.2020.9093437